

FINAL REPORT

EdTech Student Performance and Engagement Portal

Track: Education Data Systems

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Introduction

The EdTech Student Performance and Engagement Portal is a centralized educational analytics platform developed to monitor, evaluate, and enhance student activities in digital learning environments. Educational institutions generate vast amounts of data through online classes, assessments, assignments, attendance records, and learning management systems. However, this information is often distributed across multiple platforms, making it difficult for faculty members and administrators to gain meaningful insights. The proposed portal addresses this challenge by collecting, organizing, and integrating data into a unified system that enables efficient academic monitoring and informed decision-making. Through advanced data analysis and visualization techniques, the platform transforms raw educational data into actionable information that supports both teaching and learning processes.

The portal provides interactive dashboards, performance scorecards, and key performance indicators that allow students, faculty members, and administrators to track academic progress and engagement levels effectively. Automated alerts help identify students who may require additional academic support, enabling timely intervention and improved learning outcomes. The system also incorporates privacy-preserving mechanisms to ensure the security and confidentiality of sensitive educational data. By presenting information in a clear and accessible manner, the portal promotes data-driven decision-making, enhances student engagement, improves institutional effectiveness, and contributes to the overall quality of education.

Problem Statement

Educational institutions face increasing challenges in managing and analyzing the large volumes of data generated through online learning platforms, including attendance records, assessment results, assignment submissions, and student engagement activities. Since this information is often distributed across multiple systems, obtaining a comprehensive view of student performance becomes difficult. Manual monitoring methods are time-consuming, prone to errors, and may not provide timely insights for effective academic intervention. Additionally, data quality issues such as duplicate records, missing values, and inconsistencies further complicate the analysis process, limiting the ability of educators and administrators to make informed decisions.

Another major challenge is the early identification of students who may be at risk of poor academic performance or declining engagement. Traditional evaluation methods often fail

to provide real-time indicators, reducing opportunities for timely support and corrective action. Furthermore, institutions require secure and user-friendly systems that protect sensitive student information while presenting complex data through clear dashboards, reports, and performance indicators. The absence of a centralized analytics platform restricts effective educational monitoring and decision-making. Therefore, the proposed EdTech Student Performance and Engagement Portal aims to provide a unified, secure, and data-driven solution for collecting, analyzing, and visualizing educational data to improve student outcomes and institutional effectiveness.